



Predicting Failure Before It Becomes Expensive



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Stanford
University

Berkeley

SKYDECK

UC San Diego
CENTER FOR NOVEL THERAPEUTICS

HCMELAB™
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Google for Startups

One Program Forced Us to Build a Platform

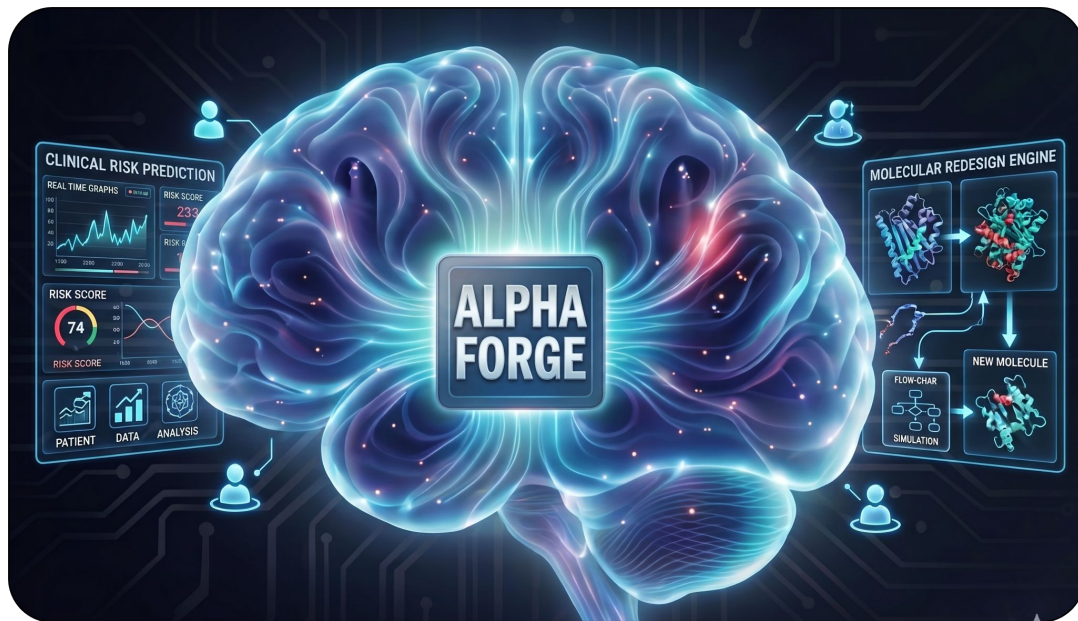
- Started as a therapeutics company
- Needed to spot risk earlier
- Built an internal AI prediction engine
- Realized it could scale far beyond one asset



Drug Failure is a Design Failure

- 
- Many failures are driven by risks encoded in molecules early
 - Those risks are usually discovered too late
 - The industry spends years & millions optimizing compounds that were never viable

Our AI Predicts Failure, Explains Why, & Redesigns Earlier

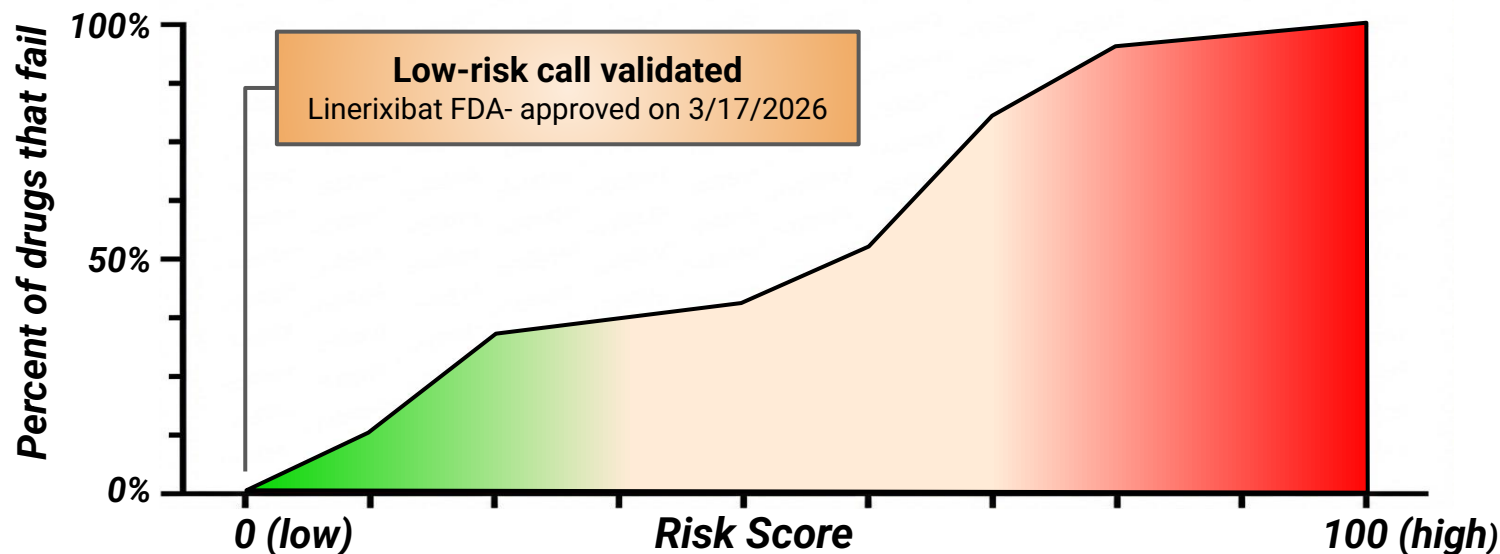


Score risk | Flag liabilities | Design lower-risk molecules

Our Risk Score Tracks Real Clinical Outcomes

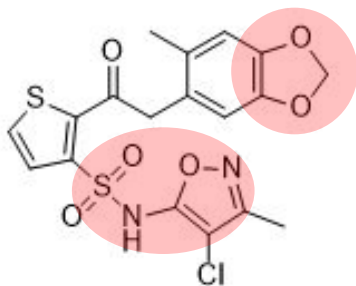
Higher score = higher failure risk

393 compounds | outcomes not used for training | AUC 0.884



AlphaForge Explains Drug Failure & Identifies Primary Risk Factors in ~80% of Failures

Bad chemistry



SITAXENTAN

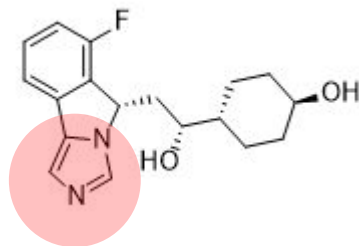
Risk Score: 95 (Very High Risk)

Platform flags **structural liabilities**:

- ❑ Benzodioxole → CYP-deactivation risk
- ❑ Sulfonamide/isoxazole → hypersensitivity risk
- ❑ Combined metabolic liability → elevated liver tox risk

Clinical Status: Approved; later withdrawn due to severe liver toxicity risk

Good chemistry, limited biology

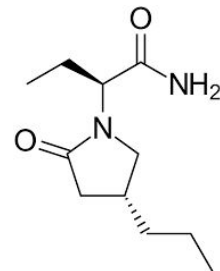


GDC-0919

Risk Score: 76 (High Risk)

- ❑ *MedChem*: No major structural liabilities.
- ❑ Azole (CYP inhibition)
- ❑ *Biology*: IDO1 activity can be compensated, limited efficacy potential

Clinical Status: Discontinued due to lack of efficacy



BRIVARACETAM

Risk Score: 5 (Low Risk)

- ❑ Balance physicochemical properties
- ❑ No major structural liabilities identified

Clinical Status: Approved CNS Drug

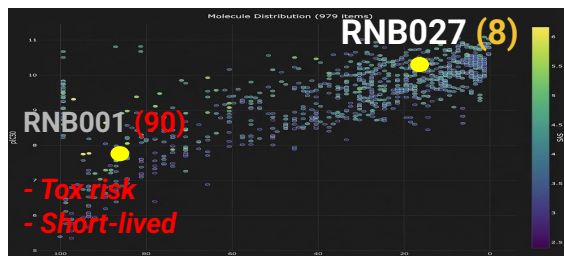
We Used AlphaForge to Rescue Our Own Program



Scored Internal program

- RNB001 (3rd Gen Academic Lead)
- Platform Score: **90** (High Risk)
- Drives regeneration & lowers inflammation
- Improves cognition

Flagged liabilities

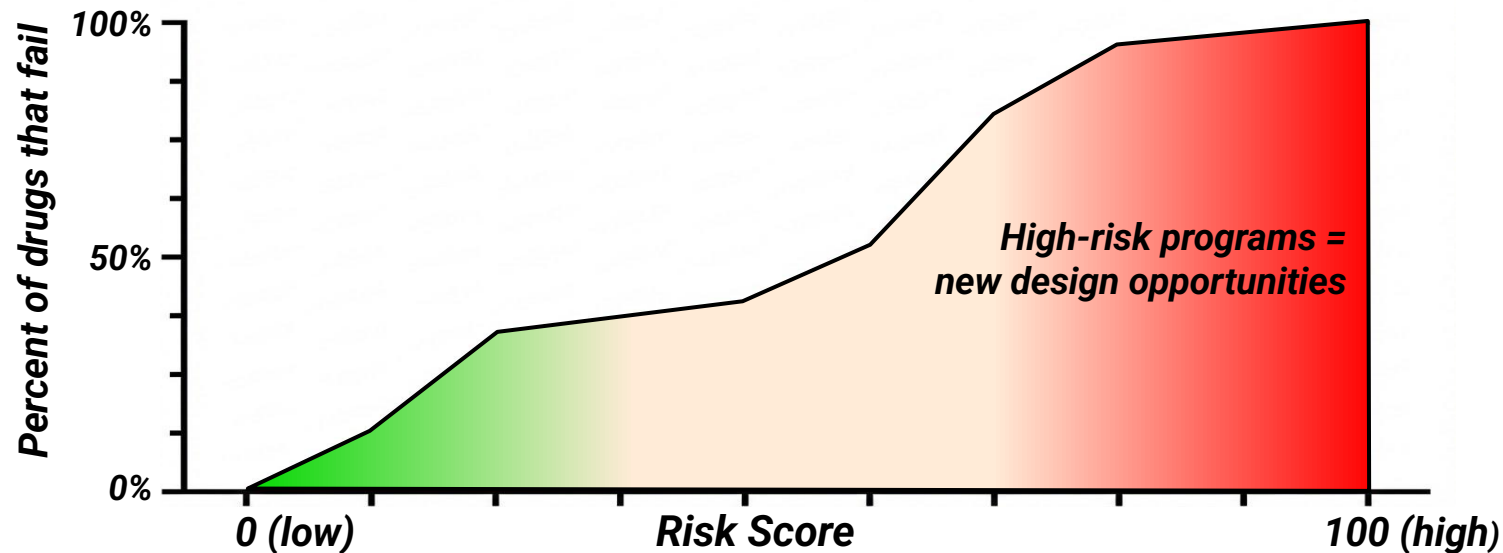


Designed RNB027 (lower-risk)

AlphaForge redesign

- Risk score: **90** → **8**
- Performance: **3x PD improvement**
- Speed: **<2 months**
- Cost: **< \$50K**

From Internal Proof to Pipeline Expansion



Ongoing collaboration with UC Berkeley's Molecular  mti
Therapeutics Initiative to apply our platform to their programs

Same Engine. First Prove Demand, Then Selectively Own Upside.



Built by Leaders in AI Drug Design, Medicinal Chemistry, & CNS Translation

Founding Team



Salah Mahmoudi, PhD
Co-founder, CEO | Drug discovery
& company-building



Tyson Ruetz, PhD
Co-founder, CSO | Aging brain
biology & target discovery



Stylianos Kyriacou, PhD
Co-founder, CTO | AI platform &
molecular design



Ramil Baiazitov, PhD
Head of Medicinal Chemistry | 19 years
in pharma; 3 assets to clinic, 1 approved



Selected Advisors/Collaborators



Anne Brunet, PhD
World-Leading Expert in Aging Biology |
Professor of Genetics | Stanford University



Julia Schaletzky, PhD
Executive Director, Molecular
Therapeutics Initiative | UC Berkeley

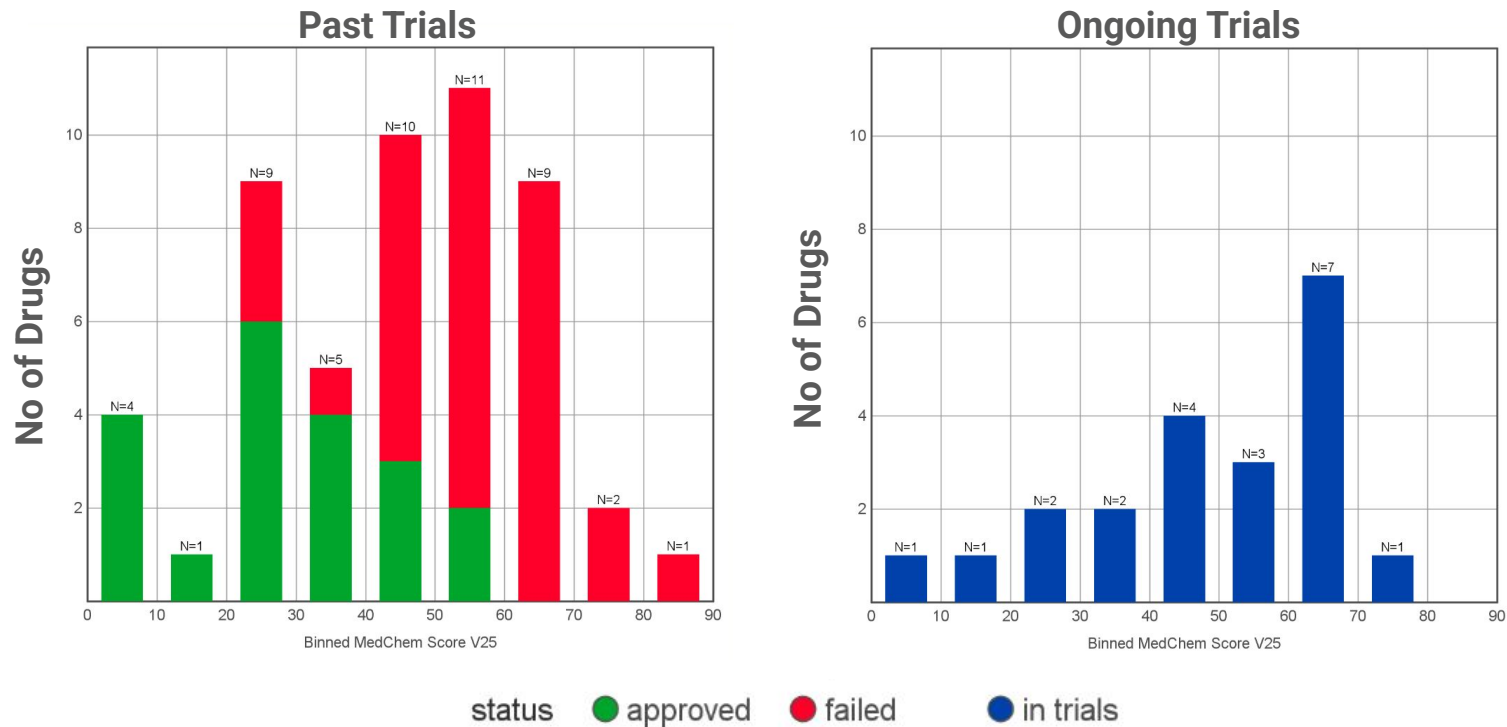


Rohan Gandhi, PhD
Clinical Translational & BD Strategy |
Autobahn Tx / Vertex

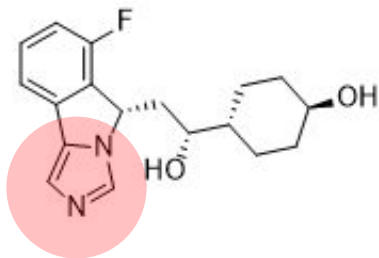


Appendix

Past & Ongoing Genentech Clinical Programs



AlphaForge Explains Why a Genentech Drugs Failed & Succeeded

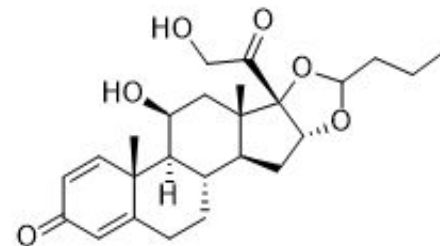


GDC-0919

Risk Score: 76 (High Risk)

- ❑ *MedChem*: No major structural liabilities.
 - ❑ Azole (potential CYP inhibition)
- ❑ *Biology*: IDO1 activity can be compensated, limited efficacy potential

Clinical Status: Discontinued due to lack of efficacy



Pulmicort

Risk Score: 1 (Very Low Risk)

- ❑ *MedChem*: No major structural liabilities identified.
- ❑ *Tox*: potential chronic safety concern, cumulative on target risks

Clinical Status: Approved